



Estimation of Pitch from Speech Signals Using Autocorrelation Algorithm

[Answer this question](#) **JAMES SMITH** 4 Jan 2018 1 Answer

Updated 23 May 2022

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Hello,

I want to detect speech signals pitch frequency using autocorrelation algorithm. I have a MATLAB code but the results are wrong. I would be grateful if you could solve the mistake in my code.

```
[y,Fs]=audioread('Sample1.wav');  
y=y(:,1);  
auto_corr_y=xcorr(y);  
subplot(2,1,1),plot(y)  
subplot(2,1,2),plot(auto_corr_y)  
[pks,locs] = findpeaks(auto_corr_y);  
[mm,peak1_ind]=max(pks);  
period=locs(peak1_ind+1)-locs(peak1_ind);  
pitch_Hz=Fs/period
```

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Thank you for your help in this matter.

2 Comments

Samia Dilbar on 14 Dec 2021

Hey did you resolve it ? help me in this

Nene Pfupa on 10 May 2022

did you find the answer.help me too.

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Asked:

JAMES SMITH

on 4 Jan 2018

Commented:

Mathieu NOE Level 8 MVP

on 23 May 2022

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hello

this is a modified version of the code . The autocorrelation output will have a major peak at sample = M (x axis) and is symmetrical around this value. So first modification is to keep the autocorr vector second half only, starting at sample = M ; what we are looking for is the next peak to get the pitch value

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```
[y,Fs]=audioread('Sample1.wav');
y=y(:,1);
auto_corr_y=xcorr(y);
M = length(y);
% C = XCORR(A), where A is a length M vector, returns the length
% auto-correlation sequence C. The zeroth lag of the output corr
% is in the middle of the sequence, at element M.

auto_corr_y = auto_corr_y(M:M+100); % keep only the second half (s
auto_corr_x = 1:numel(auto_corr_y);

[pks,locs] = findpeaks(auto_corr_y);
subplot(2,1,1),plot(y)
subplot(2,1,2),plot(auto_corr_x,auto_corr_y,locs,pks,'dr')
pitch_Hz=Fs/locs(1)
```

2 Comments



Vanya Meegan on 21 May 2022



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error unrecognized function or variable numel



Mathieu NOE Level 8 MVP on 23 May 2022



hello

numel was introduced quite recently

you can replace it with length for older matlab releases.

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y=y(:,1);
auto_corr_y=xcorr(y);
M = length(y);
% C = XCORR(A), where A is a length M vector, returns the
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auto_corr_y = auto_corr_y(M:M+100); % keep only the second l
auto_corr_x = 1:length(auto_corr_y);
[pks,locs] = findpeaks(auto_corr_y);
subplot(2,1,1),plot(y)
subplot(2,1,2),plot(auto_corr_x,auto_corr_y,locs,pks,'dr')
```

```
pitch_Hz=Fs/locs(1)
```

NS

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